

TUGAS GRAFIKA KOMPUTER

Nama: Nurhidayah Ramadhani

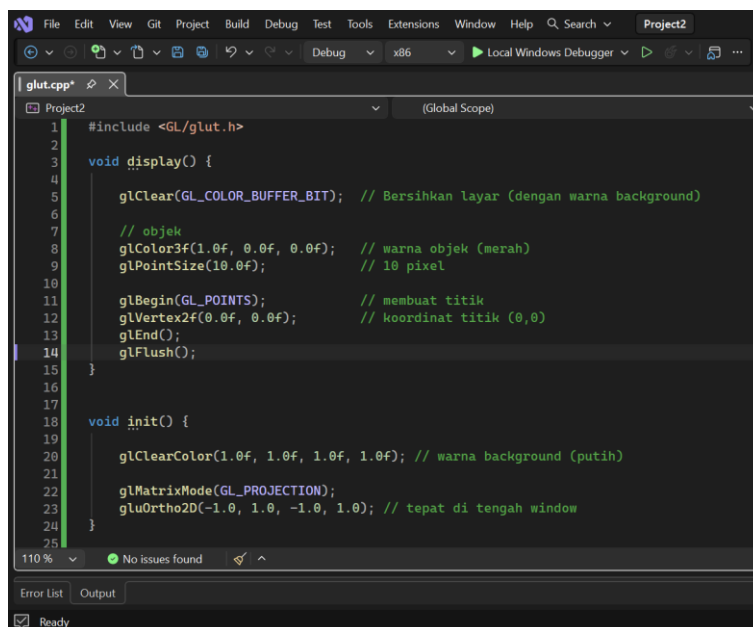
NIM: 240211060076

Soal

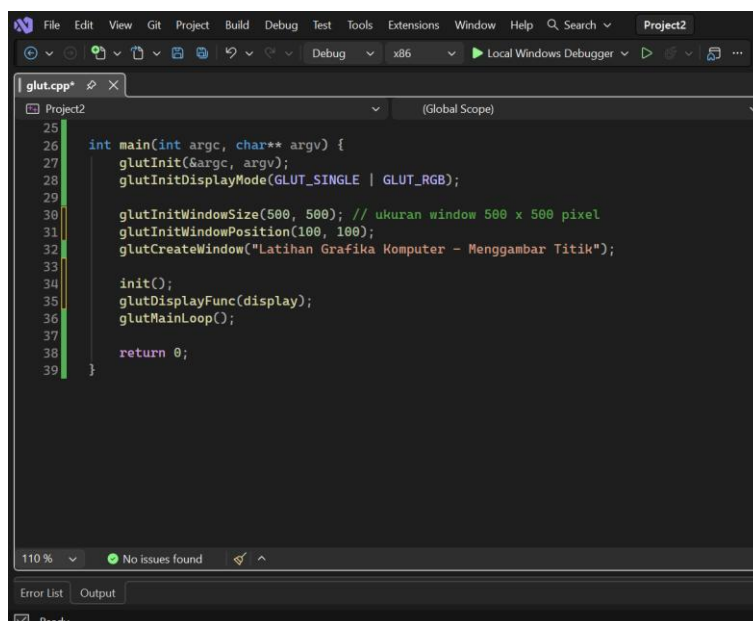
Latihan membuat titik dan garis menggunakan GLUT, submit screnshoot hasil coding dan running.

Jawaban

1. Membuat titik
 - Screnshoot hasil coding

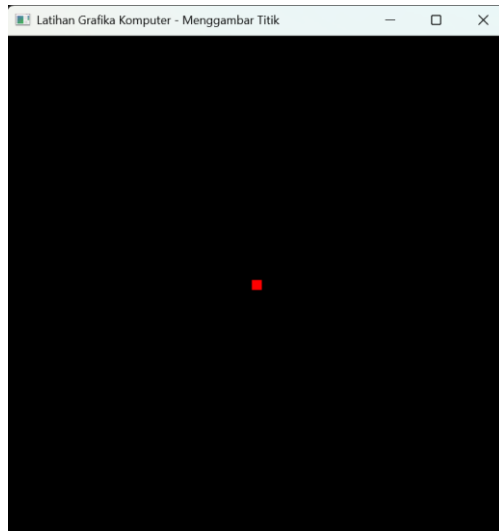


```
1 #include <GL/glut.h>
2
3 void display() {
4     glClear(GL_COLOR_BUFFER_BIT); // Bersihkan layar (dengan warna background)
5
6     // objek
7     glColor3f(1.0f, 0.0f, 0.0f); // warna objek (merah)
8     glPointSize(10.0f); // 10 pixel
9
10    glBegin(GL_POINTS); // membuat titik
11    glVertex2f(0.0f, 0.0f); // koordinat titik (0,0)
12    glEnd();
13    glFlush();
14 }
15
16
17 void init() {
18     glClearColor(1.0f, 1.0f, 1.0f, 1.0f); // warna background (putih)
19
20     glMatrixMode(GL_PROJECTION);
21     gluOrtho2D(-1.0, 1.0, -1.0, 1.0); // tepat di tengah window
22 }
23
24
25
```



```
25 int main(int argc, char** argv) {
26     glutInit(&argc, argv);
27     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
28
29     glutInitWindowSize(500, 500); // ukuran window 500 x 500 pixel
30     glutInitWindowPosition(100, 100);
31     glutCreateWindow("Latihan Grafika Komputer - Menggambar Titik");
32
33     init();
34     glutDisplayFunc(display);
35     glutMainLoop();
36
37     return 0;
38 }
39
```

- Screenshoot hasil running



2. Membuat garis

- Screenshoot hasil coding

```
1 #include <GL/glut.h>
2
3 void display() {
4     glClear(GL_COLOR_BUFFER_BIT);
5     glColor3f(1.0f, 0.75f, 0.8f); // Objek warna pink
6
7     glLineWidth(3.0f); // Ketebalan garis
8
9     glBegin(GL_LINES); // Membuat garis
10    glVertex2f(-0.8f, 0.0f); // titik awal (x1, y1) -> y = 0 (tengah)
11    glVertex2f(0.8f, 0.0f); // titik akhir (x2, y2) -> y = 0 (tengah)
12    glEnd();
13    glFlush();
14 }
15
16 void init() {
17     glClearColor(0.0f, 0.0f, 0.0f, 1.0f); // background hitam
18
19     glMatrixMode(GL_PROJECTION);
20     gluOrtho2D(-1.0, 1.0, -1.0, 1.0); // area koordinat -1 s/d 1
21 }
22
23 int main(int argc, char** argv) {
```

```
24     glutInit(&argc, argv);
25     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
26     glutInitWindowSize(500, 500);
27     glutInitWindowPosition(100, 100);
28     glutCreateWindow("Latihan Grafika Komputer - Menggambar Garis");
29
30     init();
31     glutDisplayFunc(display);
32     glutMainLoop();
33
34     return 0;
35 }
```

- Screenshot hasil running

